Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/14/2018
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Mechanisms of CDKN1a/p21 Inhibition of Tissue Regeneration in Microgravity - Roles of Mechanical Unloading and Oxidative Stress		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	12/02/2013	**End Date:**	12/01/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
Contact Email:	npphelp@usra.edu		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo Ph.D. (MENTOR/ NASA Ames Research Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
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Performance Goal Text:			
	NASA Postdoctoral Program Fellow Tissue degeneration and the failure of normal stem cell-based tissue regeneration in microgravity pose significant risks for astronaut health during long-duration spaceflight missions. The immediate degenerative effects we understand best are those studied during short-term Low-Earth orbit spaceflight experiments and include rapid microgravity adaptive bone and muscle loss, loss of cardiovascular capacity, defects in wound and bone fracture healing, and impaired immune function. Although bone loss in space has been studied in both humans and rodent models, such as mice, the precise mechanisms by which bone loss occurs are still currently debated. Rapid bone loss has been well documented within the first couple of days of spaceflight and has been attributed to increased osteoclast resorption of trabecular bone; however, this effect seems to be a rapid and short term response to unloading. Perhaps of more relevance to long		

Task Description:	duration spaceflight missions is the possible slowing or alteration in bone growth that occurs in microgravity, characterized by a decrease in the osteoblast population of both cortical and trabecular bone. Furthermore, osteoblasts have been shown to exhibit altered cell growth and proliferation, further indicating a role for loss of bone growth in addition to osteoclast-mediated bone degradation in spaceflight-induced osteopenia. During our previous short-duration (15 days) spaceflight studies with mice, we have addressed the problem of regenerative deficits in microgravity, with a specific focus on bone tissue and bone marrow stem cells. We have found that microgravity impairs the ability of mesenchymal and hematopoietic stem cells to differentiate into terminally differentiated cells, including osteoblasts, osteoclasts, and chondrocytes. Furthermore, we have investigated the impact of altered somatic stem cell regenerative capacity on bone tissue, and found significant degeneration of bone tissue due to both increased bone resorption and decreased regenerative bone formation. Through these studies, we also identified a significant molecular mechanism, activation of the cell cycle arrest molecule CDKN1a/p21, in proliferating osteoprogenitors on the bone surface and proliferating chondrocytes that may account for the observed alterations in tissue regeneration in space. Global gene expression studies of bone and bone marrow isolated from space-flown mice also showed significant alterations in proliferation and differentiation pathways. Altered gene expression included several genes highly relevant to regenerative health including cold inducible RMNA binding protein, DNA damage inducible transcript 4 (DDIT4), and interleukins IL1ß and IL1F9. Furthermore, we found down-regulation of a master controller for oxidative stress mitigation in both bone tissue and bone marrow, NFE2L2 (Nrf2), suggesting increased levels and susceptibility to reactive oxygen species. Taken together, this data indicated that CDKN1a/p21 could be an important mechanism for the regulation of tissue regeneration in response to alterations in mechanical load. As CDKN1a/p21 can be induced by oxidative stress, and as mechanical unloading is also thought to induce oxidative stress, we hypothesize that oxidative stress induced by mechanical unloading could induce p21 over-expression during spaceflight resulting in the observed inhibition of regenerative bone formation in space. Therefore, the aims of this proposal are to firstly build upon our previous spaceflight results by using the CDKN1a/p21 knockout (KO) mouse to investigate the role of CDKN1a/p21 in regenerative arrest induced by ground-based models of microgravity (hindlimb unloading) and secondly, to elucidate the molecular mechanism connecting mechanical unloading to oxidative stress and CDKN1a/p21 induction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	To achieve the above, we firstly aimed to conduct a hindlimb unloading experiment with KO (knock out) mice and test the hypothesis the CDKN1a/p21 pathway is part of the molecular mechanism for microgravity inhibition of mesenchymal stem cell-mediated bone tissue regeneration in-vivo. We also aimed to conduct a uniaxial mechanical load test with CDKN1a/p21 KO stromal mesenchymal stem cell cultures to test the hypothesis that mechano-stimulation may directly down-regulate CDKN1a/p21 levels in stem cells undergoing regenerative differentiation through either transcriptional or translational mechanisms. Finally, we aimed to test the hypothesis that the CDKN1a/p21 pathway in osteoprogenitors is regulated by oxidative stress in-vivo. During my NASA Postdoctoral Program (NPP) appointment, we made significant progress on all aims of the above research plan. Additionally, we wrote and were awarded a NASA spaceflight grant to investigate the influence of microgravity on osteoprogenitor differentiation into mineralized nodules on the International Space Station (ISS) using the Bioculture System hardware, enabling us to specifically compare and contrast ground-based differentiation under conditions of increased mechanical stimulation. Furthermore, we have published 4 articles in peer-reviewed journals (3 research articles and 1 review article) based on our previous spaceflight experiments showing inhibition of stem cell regenerative capacity in space and activation of CDKN1a/p21 as a potential molecular mechanism causing this inhibition. These papers pave the way and lay foundations for the research articles that will be published based on the specific role of CDKN1a/p21 on regenerative bone formation. Overall, these studies have significantly contributed to our knowledge of how mechanical unloading regulates stem cell-based tissue regeneration and the role that CDKN1a/p21 plays in this process. The spaceflight grants that were awarded during this NPP fellowship will enable us to further elucidate the specific role of CDKN1a/p21 on mesenchymal stem cell proliferation, differentiation and regeneration capacities during microgravity exposure on the ISS.
Bibliography Type:	Description: (Last Updated: 04/03/2025)
Articles in Peer-reviewed Journals	Blaber E, Sato K, Almeida EA. "Stem cell health and tissue regeneration in microgravity." Stem Cells Dev. 2014 Dec;23 Suppl 1:73-8. https://doi.org/10.1089/scd.2014.0408 ; PubMed PMID: 25457968; PubMed Central PMCID: PMC4235978 , Dec-2014
Articles in Peer-reviewed Journals	Blaber EA, Finkelstein H, Dvorochkin N, Sato KY, Yousuf R, Burns BP, Globus RK, Almeida E. "Microgravity reduces the differentiation and regenerative potential of embryonic stem cells." Stem Cells Dev. 2015 Nov 15;24(22):2605-21. Epub 2015 Oct 22. https://doi.org/10.1089/scd.2015.0218 ; PubMed PMID: 26414276; PubMed Central PMCID: PMC4652210 , Nov-2015
Journal/Magazine covers	Blaber EA, Finkelstein H, Dvorochkin N, Sato KY, Yousuf R, Burns BP, Globus RK, Almeida E. "Cover in journal Stem Cells and Development for article, 'Microgravity reduces the differentiation and regenerative potential of embryonic stem cells.' " Stem Cells Dev. 2015 Nov 15;24(22):2605-21. PubMedPMID: 26414276; PubMed Central PMCID: PMC4652210 ; https://www.liebertpub.com/doi/10.1089/scd.2015.0218 , Nov-2015